

ReconX

Enterprise Trade Reconciliation Platform

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The Problem

- Ops teams reconcile thousands of trades a day against custodian and counterparty feeds — largely by hand
- Mismatches surface late, after settlement risk has already built up, and cost real money to unwind
- ReconX auto-detects breaks in near-real-time so Ops resolves them, not discovers them

Why it matters

A single undetected break can sit unresolved for a full settlement cycle. Automated, streaming reconciliation turns a next-day discovery into a same-minute alert.

Architecture — C4 Container View

Recon UI

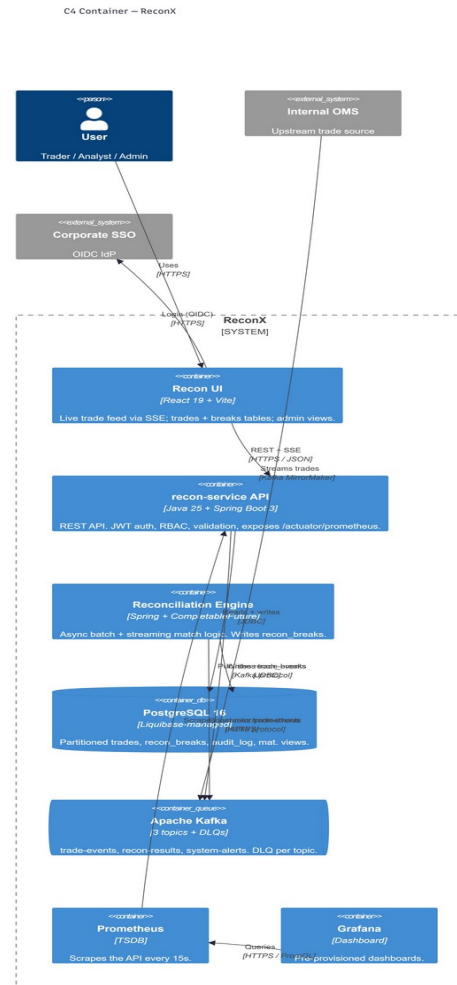
React 19 + Vite, live SSE feed

recon-service API

Java 25 + Spring Boot 3, JWT + RBAC

Reconciliation Engine

Async match + break detection



PostgreSQL 16

Liquibase-managed schema

Apache Kafka

3 topics + DLQ per topic

Prometheus + Grafana

Scrapes /actuator/prometheus

Tech Stack — by Layer

Data

PostgreSQL 16 · Liquibase migrations

Application

Java 25 · Spring Boot 3 · Spring Security (JWT/RBAC)

Messaging

Apache Kafka · 3 topics · DLQ per topic

UI

React 19 · Vite · Server-Sent Events

Observability

Prometheus · Grafana

Live Demo — Login & Post a Trade

- Log in as trader@db.com — JWT issued, role-checked
- Submit a new trade from the Recon UI
- Request validated, persisted, and published to Kafka

Fallback

Screen recording from rehearsal, pending capture (ADV163).

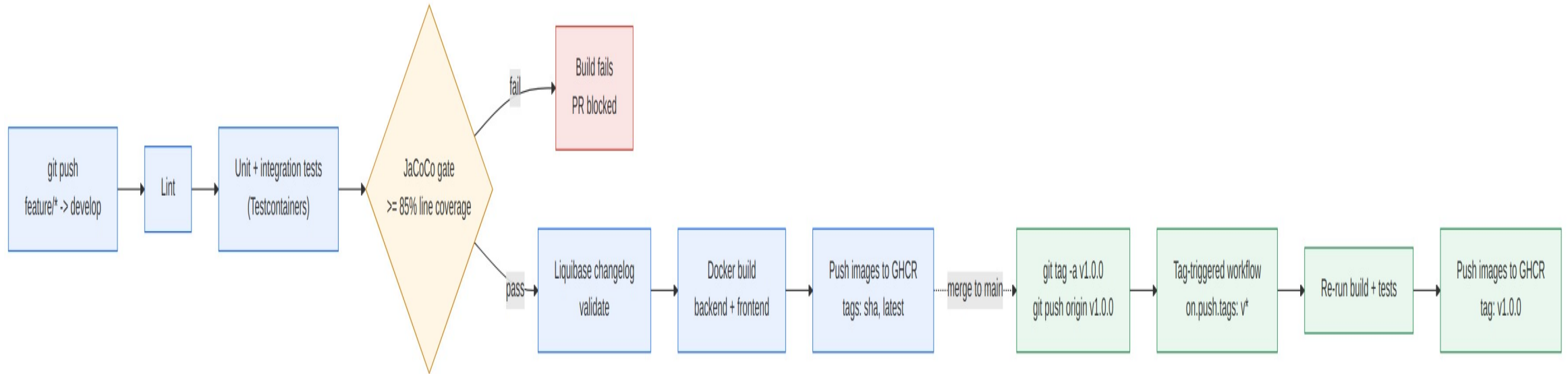
Live Demo — Kafka & Auto-Reconciliation

- Trade lands on the trade-events topic (Kafdrop)
- Reconciliation engine consumes it and runs match logic
- Grafana's request-rate panel ticks up in real time

Fallback

Kafdrop + Grafana screenshot, pending capture (ADV159).

CI/CD Pipeline



85% JaCoCo coverage gate

Liquibase validate · Docker build · GHCR push on green main · tag-triggered v1.0.0 rebuild

Monitoring

- Prometheus scrapes /actuator/prometheus every 15s
- Grafana dashboards pre-provisioned (request rate, latency, break count)
- Screenshots skipped for this cohort per instructor guidance

Scope note

Grafana screenshots (ADV159) were explicitly marked optional/skip by the instructor for this cohort — not an open item.

Learnings

Avani

Lead

— *one-sentence takeaway (spoken live)*

Vigna

Backend — API

— *one-sentence takeaway (spoken live)*

Nandini

Backend — Kafka

— *one-sentence takeaway (spoken live)*

Shubhang

Frontend

— *one-sentence takeaway (spoken live)*

Gaurav

DevOps / CI

— *one-sentence takeaway (spoken live)*

Questions?

github.com/<org>/reconx

Avani routes questions — "we didn't get to that" is a valid answer.